

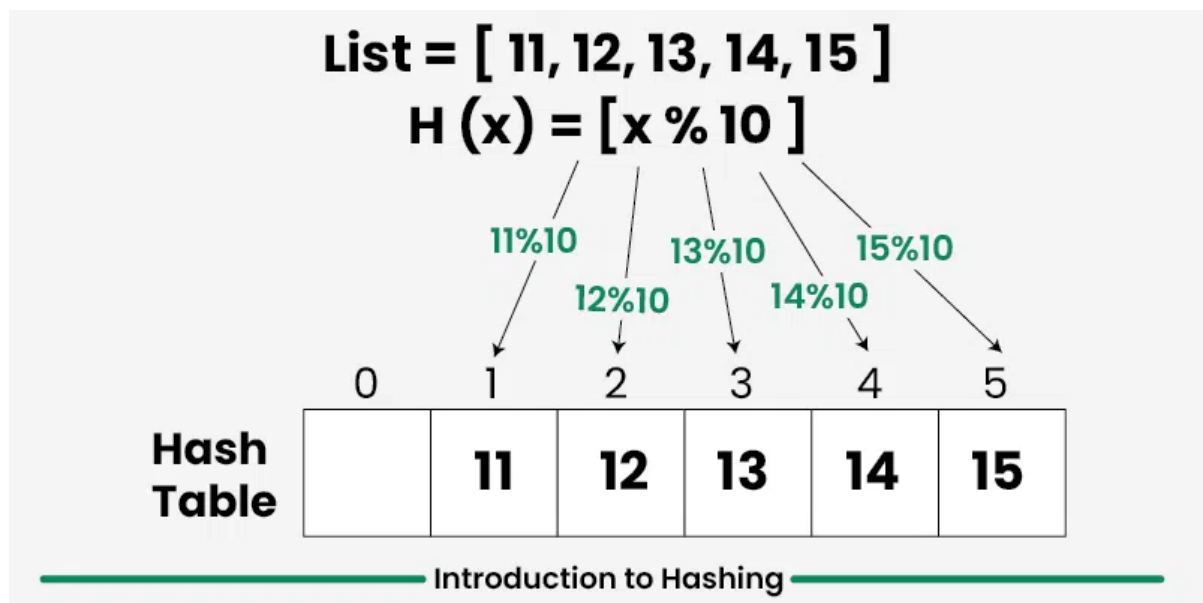
Hashing in Data Structure

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Hashing is a technique used in data structures that efficiently stores and retrieves data in a way that allows for quick access.

- Hashing involves mapping data to a specific index in a hash table (an array of items) using a **hash function**. It enables fast retrieval of information based on its key.
- The great thing about hashing is, we can achieve all three operations (search, insert and delete) in $O(1)$ time on average.
- Hashing is mainly used to implement a set of distinct items (only keys) and dictionaries (key value pairs).



Hashing in Data Structure

Basics

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- [Applications](#)
- [Separate Chaining for Collision Handling](#)
- [Open Addressing for Collision Handling](#)

Easy Problems

- [Check for Subset](#)

- [Check for Disjoint](#)
- [Check for Equal](#)
- [Fizz Buzz](#)
- [Max distance between two occurrences](#)
- [Duplicate within K Distance](#)
- [Intersection of Two Arrays](#)
- [Union of Two Arrays](#)
- [Most Frequent Element](#)
- [Maximum points on the same line](#)
- [2 Sum - Find if there is any pair](#)
- [2 Sum - Count Pairs](#)
- [Count Pairs with Given Diff](#)
- [Only repetitive element from 1 to n-1](#)
- [Missing elements of a given range](#)
- [Missing Elements from Min to Max of Array](#)
- [Minimum Subsets with distinct elements](#)
- [Minimum Removals for No Common](#)
- [Pairs with given sum from different rows](#)

Medium Problems

- [Check If Array Pair Sums Divisible by k](#)
- [Longest subarray with sum divisible by k](#)
- [3 Sum - Count all triplets with given sum](#)
- [3 Sum – Find All Triplets with Zero Sum](#)
- [Itinerary from a given list of tickets](#)
- [Longest Subarray having Majority Elements Greater Than K](#)
- [Number of Employees Under every Employee](#)
- [Largest subarray with 0 sum](#)
- [Subarray with given sum](#)
- [Longest Consecutive subsequence](#)
- [Largest Fibonacci Subset](#)
- [Consecutive Subset Partitioning](#)
- [Distincts in every window of size k](#)
- [Insert, delete, search and getRandom](#)

- [Min insertions for a palindrome permutation](#)
- [Maximum possible difference of two subsets of an array](#)
- [Sorting using trivial hash function](#)
- [Smallest subarray with k distinct numbers](#)
- [All pairs \(a, b\) in an array such that \$a \% b = k\$](#)
- [Group words with same set of characters](#)
- [k-th distinct \(or non-repeating\) element in an array.](#)

Hard Problems

- [Represent Fraction as String](#)
- [4 Sum – Count quadruplets](#)
- [4 Sum – Find all Quadruplets](#)
- [4 Sum - From four sorted arrays](#)
- [Clone a Binary Tree with Random Pointers](#)
- [Largest subarray with equal number of 0s and 1s](#)
- [Longest Common Sum Span](#)
- [Palindrome Substring Queries](#)
- [Range Queries for Frequencies of array elements](#)
- [Cuckoo Hashing – Worst case \$O\(1\)\$ Lookup!](#)
- [Count subarrays having total distinct elements same as original array](#)
- [Maximum array from two given arrays keeping order same](#)
- [Find Sum of all unique sub-array sum for a given array.](#)
- [Recaman's sequence](#)
- [Length of longest strict bitonic subsequence](#)
- [Find All Duplicate Subtrees](#)
- [Find if there is a rectangle in binary matrix with corners as 1](#)

Quick Links :

- [‘Practice Problems’ on Hashing](#)
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Intermediate problems on Hashing

